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Faculty of Technology

Department of Information & Communications Technology

PROFESSIONAL ETHICS

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Q1.

1. Manipulative / Addictive Design (Dark Patterns)

The platform intentionally designs notifications to manipulate users' behaviour and increase screen time by triggering dopamine responses.

2. Negligence toward User Well-being

The company knowingly continued the design despite internal research showing it increases anxiety among users under 25, violating the duty to protect user welfare.

Q2

1. Informed Consent Violation

The disclosure is hidden in paragraph 47 of a 90-page document, meaning users are unlikely to understand what they agreed to.

2. Privacy Violation

Collecting precise location every 30 seconds intrudes on personal privacy.

3. Data Minimization Violation

The company collects far more data than necessary for normal fintech services.

4. Transparency Failure

Important information is hidden in lengthy legal text instead of being clearly explained.

5. Purpose Limitation Violation

Data collected for fintech services is later sold to insurance companies for another purpose.

6. Data Retention Abuse

Storing location data indefinitely increases risks of misuse or breaches.

Q3.

Ethical failure:

Ignoring environmental responsibility and sustainability.

Explanation:

The team chose a coal-powered region simply to reduce short-term costs, despite a greener alternative being available.

Responsibility:

Both engineering leadership and company management share responsibility because they prioritised budget targets over environmental impact.

Q4.

Two actions to take

1. Report the issue internally to legal/compliance or senior management.
2. Remove or properly license the GPL code and provide required attribution.

Consequences of doing nothing

- Legal consequences
- Violation of the GPL licence
- Possible lawsuits
- Forced release of the company's source code

Ethical consequences

- Intellectual property theft
- Loss of professional integrity
- Damage to company reputation

Q5.

False Positive	False Negative
The AI system incorrectly identifies a candidate as unsuitable.	The AI system incorrectly accepts an unsuitable candidate.
Example: A qualified candidate is rejected.	Example: An unqualified candidate gets selected.

Why minimizing one increases the other

If you make the system more strict, fewer unqualified candidates pass but more qualified candidates may be rejected.

Why this is ethical

Because it affects:

- fairness
- discrimination
- people's careers
- Therefore it is not purely technical.

Q6.

Why ethical violation:

- Violates responsible disclosure principle
- Puts users at risk
- Uses security flaw for business advantage

Correct action:

1. Follow responsible disclosure
2. Inform competitor privately
3. Allow time to fix
4. Then disclose publicly if needed

Q7

Ethical obligations violated:

1. Duty to ensure safety and security
2. Professional responsibility
3. Public interest priority
4. Quality assurance obligation

What developer should do:

- Report risk
- Document issue
- Inform higher authority / ethics channel
- Refuse unsafe release if possible

Q8

Two ethical failures:

1. Algorithmic Bias / Discrimination

The system performs significantly worse for dark-skinned faces (74% vs 98%).

2. Misleading Use of Overall Accuracy

The company hides inequality by focusing only on average accuracy instead of fairness across groups.

Q9.

What must you do:

- Declare conflict of interest
- Inform client about payment offer
- Stay transparent

If not:

- Loss of professional credibility
- Violation of professional ethics
- Possible legal consequences
- Loss of trust

Q10.

Ethical principle violated:

- Accessibility / Fairness / Inclusion
- Also violates *user interest priority

Harm caused:

- Users forced to buy new devices
- Excludes low-income users
- Creates digital inequality

Technical solution:

- Optimize code
- Support older OS versions
- Provide lightweight version of app

Case Study 1

Should the public have a right to know?

- Yes, users have the right to know about risks to their personal data.
- Transparency builds trust between the company and users.
- Hiding security issues may protect reputation but harms user safety.

Is the engineer ethically obligated to report it?

- Yes, according to professional ethics, protecting users is more important than protecting company image.
- The engineer should follow responsible disclosure rules.
- Reporting internally first is the correct step.

Responsible Disclosure	Whistleblowing
Report a problem privately to the company.	Report a problem publicly.
Gives the company a chance to fix it first.	Done when the company does not fix it.
Protects users and the company.	Protects the public/society.

Case Study 2

Ethical duty

- Company must ensure fair hiring.
- AI should not discriminate.
- All applicants deserve equal chance.

Stop or improve

- Improve the system.
- Use better training data.
- Add human checking.

Reduce bias

- Test with different data.
- Do regular audits.
- Check results for fairness.

Case Study 3

Ethical issue

- Yes, collecting extra data without consent is wrong.
- Users were not informed.

Data minimization & transparency

- Collect only needed data.
- Clearly tell users what is collected.

Prevention

- Clear privacy policy.
- Ask permission.
- Give user control.

Case Study 4

Should engineers push back?

- Yes, engineers should consider environmental impact.
- Professional ethics include social responsibility.
- High energy use affects sustainability.

Professional duties

- Give honest technical advice.
- Suggest efficient alternatives.
- Avoid harmful designs if possible.

Can aesthetics and sustainability coexist?

- Yes, good design can be both efficient and attractive.
- Use optimized graphics.
- Use lightweight coding.
- Balance performance and beauty.

Case Study 5

Report externally?

- Try internal reporting first.
- If illegal, external reporting may be needed.

Protection in Sri Lanka

- Some legal protection exists.
- But not always strong.

Internal steps

- Inform manager / HR.
- Keep records.
- Suggest legal solution.

Case Study 6

Responsibility

- Platform owners must set rules.
- Developers must design safe systems.
- Users must follow guidelines.

Ethical design

- Moderation tools.
- Report / block options.
- Identity checks.

Accountability

- Verified identity reduces misuse.
- Logs help track users.

Case Study 7

Moral rights

- Developer should get credit.
- Creator has rights to work.

Client responsibility

- Must respect agreement.
- Removing credit is unfair.

IP & fairness

- Contract should define ownership.
- Fair pay and credit are important.

Case Study 8

Ethical problem

- Collecting facial data without consent is wrong.
- Sharing data with companies violates privacy.

Engineer role

- Warn about risks.
- Follow ethical standards.

Public role

- People should be informed.
- Public consent is important.

Case Study 9

Exploitation

- When student does full work without pay.
- No learning benefit.
- False promises.

Institution role

- Universities should protect students.
- Internship rules needed.

Ethical policy

- Clear agreement.
- Fair work.
- Learning focus.

Case Study 10

Ownership

- Not always clear.
- Could be company, user, or artist.

Ethics

- Using art without consent is unfair.
- Creators should be respected.

Engineer duty

- Use licensed data.
- Check consent.
- Follow ethical AI rules.